

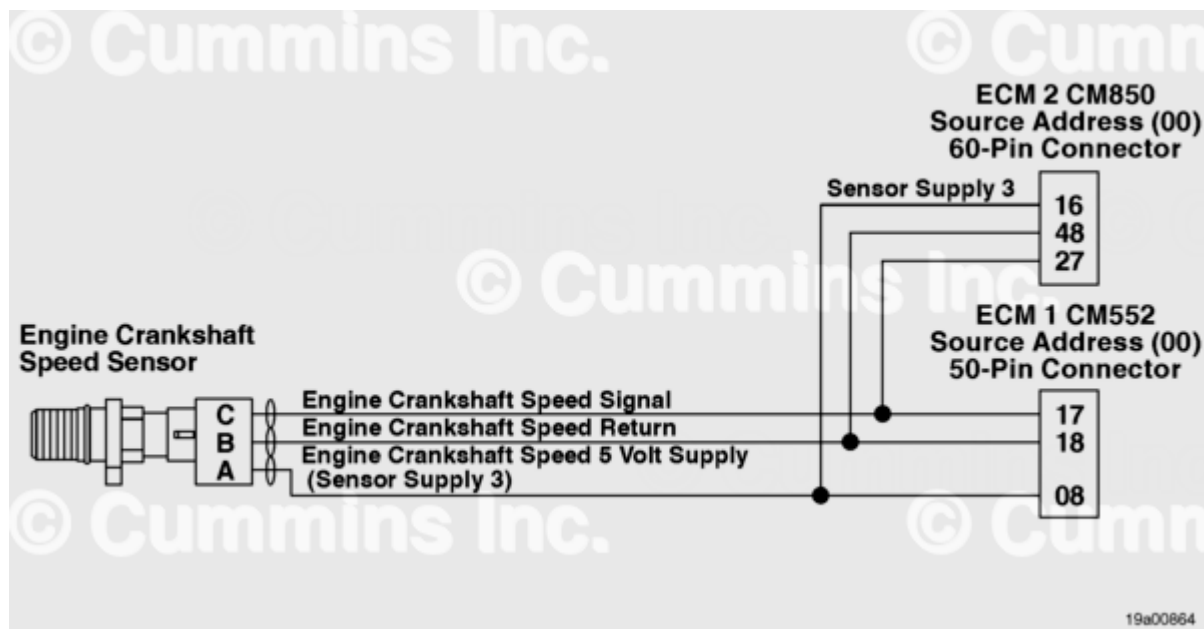
Fault Code: 1992

Engine Crankshaft Speed/Position - Data Valid But Above Normal Operating Range - Moderately Severe Level

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Overview

Codes	Reason	Effect
Fault Code: 1992 PID(P): P190 SPN: 190 FMI: 0/16 Lamp: Red SRT:	Engine crankshaft speed signal indicates engine speed above engine protection limit.	Engine will shut down.



Engine Crankshaft Speed Sensor Circuit

Circuit Description

The crankshaft position and camshaft position sensors are Hall effect type sensors. The engine control module (ECM) provides a 5 volt supply to the position sensor and a return circuit. As the teeth on the crankshaft speed ring or the dimples in the back of the camshaft gear move past the

position sensor, a signal is generated on the position sensor signal circuit. The ECM interprets this signal and converts it to an engine speed. A missing tooth on the crankshaft gear is used by the ECM to determine the position of the engine.

Component Location

The engine crankshaft speed/position sensor is located on the flywheel housing.

Conditions For Running The Diagnostics

This diagnostic runs continuously when the engine is running.

Conditions For Setting The Fault Codes

The Engine Control Module (ECM) detected the engine crankshaft speed sensor is higher than the engine protection limits.

Action Taken When The Fault Code Is Active

- The generator set controller displays a shutdown fault immediately when the diagnostics runs and fails.
- The engine will be shut down.
- Fueling to the engine is stopped until the engine speed drops to normal operating speeds.

Conditions For Clearing The Fault Code

- To validate the repair, start the engine and let it run for 1 minute at no load.
- The generator set controller will turn off the shutdown indicator immediately after the user presses the reset.
- The "Reset All Faults" command in the recommended Cummins® electronic service tool or equivalent can be used to clear active faults.

Shoptalk

There are multiple ECMs for the engine models included in this manual. Each ECM has an individual source address that displays when Cummins® electronic service tool or equivalent is connected. When troubleshooting a fault code, use the source address displayed in Cummins® electronic service tool or equivalent to determine which ECM and circuit is affected.

Possible causes of this fault code include:

- External fuel sources drawn into the intake air passage
- Reverse powering (motoring) of the engine

- Tampering of the engine speed/position sensors.

Refer to Troubleshooting Fault Code 1992 . (</qs3/pubsys2/xml/en/procedures/60/60-t05-1992.html>)

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